THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The first 8 odd numbers add up to: 1

(a) 65 **(b)** 56 **(c)** 64 **(d)** 72

2. The next number in the pattern $10, 15, 20, 25, 30, \ldots$ is: 1

(a) 36 **(b)** 35 **(c)** 34 **(d)** 37

3. The 20th odd number is: 1

(a) 38 **(b)** 39 **(c)** 41 **(d)** 40

4. The 18th even number is: 1

(a) 35 **(b)** 36 **(c)** 38 **(d)** 37

5. The value of 13^2 (the 13th square number) is: 1

(a) 169 **(b)** 170 **(c)** 182 **(d)** 156

6. The 7th triangular number $1 + 2 + \cdots + 7$ is: 1

(a) 21 **(b)** 29 **(c)** 28 **(d)** 35

7. The value of 5^3 (the 5th cube number) is: 1

(a) 150 **(b)** 130 **(c)** 125 **(d)** 120

8. The first 9 odd numbers add up to: 1

(a) 82 **(b)** 90 **(c)** 81 **(d)** 72

9. The next number in the pattern $3, 9, 15, 21, 27, \ldots$ is: 1

(a) 34 **(b)** 35 **(c)** 32 **(d)** 33

10. The 25th odd number is: 1

(a) 49 **(b)** 48 **(c)** 50 **(d)** 51

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the first 7 triangular numbers. 2

12. Write the next three numbers: $15, 19, 23, 27, \ldots$ 2

13. Write the first 8 square numbers. 2

14. Write the first 8 triangular numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the 8th term of the pattern $6, 9, 12, 15, \ldots$ by adding the common difference repeatedly. 3

16. Show that the sum of the first 10 odd numbers is a square number. 3

17. Write the numbers 1 to 13 as a growing sum and find the 13th triangular number. 3

OR

Find the 7th term of the pattern $1, 7, 13, 19, \ldots$ by adding the common difference repeatedly. 3

18. Show that the sum of the first 12 odd numbers is a square number. 3

OR

Write the numbers 1 to 14 as a growing sum and find the 14th triangular number. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the first 6 terms of the pattern which starts at 3 and adds 8 each time. Is the number 59 in this pattern? Which term is it? 5

OR

Write 64 as the sum of consecutive odd numbers starting from 1, and write it as a product of two equal numbers. Then write the sum of the first 9 odd numbers. 5

20. Write the first 6 terms of the pattern which starts at 5 and adds 6 each time. Is the number 47 in this pattern? Which term is it? 5

OR

Write 81 as the sum of consecutive odd numbers starting from 1, and write it as a product of two equal numbers. Then write the sum of the first 10 odd numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND